

power is available, diode D5 forward biases. It provides power to the circuit and also charges the battery through resistor R2, and it limits the charging current to 120 mA. When power fails, diode D5 reverse biases and diode D6 forward biases, giving instant backup

to the circuit. LED1 indicates the availability of mains power.

Assemble the circuit on a general-purpose PCB and enclose it in a suitable case. Connect the piezo element to the circuit using a thin insulated wire. Glue the flat side of the piezo el-

ement on a 30×30cm aluminium sheet to increase its sensitivity. Fix the sheet with the piezo sensor to the site where protection is needed. The remaining circuit can be fixed at a suitable place. If only the alarm generator is needed, omit the relay driver section. ●

TELEPHONE CALL RECORDER

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Today telephone has become an integral part of our lives. It is the most widely used communication device in the world. Owing to its immense popularity and

widespread use, there arises a need for call recording devices, which find application in call centres, stock broking firms, police, offices, homes, etc.

Here we are describing a call recorder that uses very few components. But in order to understand its working,

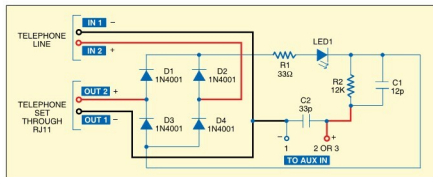


Fig. 1: Call recorder circuit



Fig. 2: Pin configuration of stereo jack

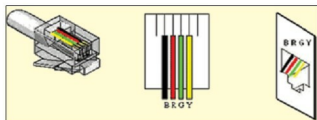


Fig. 3: RJ connector

one must first have the basic knowledge of standard telephone wiring and a stereo plug.

In India, landline telephones primarily use RJ11 wiring, which has two wires—tip and ring. While tip is the positive wire, ring is the negative one. And together they complete the telephone circuit. In a telephone line, voltage between tip and ring is around 48V DC when handset is on the cradle

(idle line). In order to ring the phone for an incoming call, a 20Hz AC current of around 90V is superimposed over the DC voltage already present in the idle line.

The negative wire from the phone line goes to IN1, while the positive wire goes to IN2. Further, the negative wire from OUT1 and the positive wire from OUT2 are connected to the phone. All the resistors used are 0.25W carbon film resistors and all the capacitors used are rated for 250V or more. The negative terminal of 'To AUX IN' is connected to pin 1 of the stereo jack while the positive terminal is connected to pins 2 and 3 of the stereo jack. This stereo jack, in turn, is connected to the AUX IN of any recording device, such as computer, audio cassette player, CD player, DVD player, etc. Here we shall be connecting it to a computer.

When a call comes in, around 90V AC current at 20Hz is superimposed over the DC voltage already present in the idle line. This current is converted into DC by the diodes and fed to resistor R1, which reduces its magnitude and feeds it to LED1. The current is further reduced in magnitude by the resistor R2 and fed to the right and left channels of the stereo jack, which are connected to the AUX IN port of a computer.

Any audio recording software, such as AVS audio recorder (available at: <http://www.avs4you.com/AVS-Audio-Recorder.aspx>), Audacity audio recorder (<http://audacity.sourceforge.net/>), or audio recorder (http://www.audio-tool.net/audio_recorder_for_free.html), can be used to record the call. When a call comes in, one needs

